Maria Paola Vaccaro

Born on July 11th, 1997
in Larino (CB), Italy
Currently in Heidelberg, Germany

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Scientific interests

- Computational astrophysics
- Compact objects
- Stellar evolution
- Gravitational waves sources
- Black holes dynamics
- Stellar clusters

Numerical tools

Python	●●●●
C	●●●○
Matlab	●●●○

Operating systems

Linux/Unix	●●●○
Windows	●●●○

Languages

Italian (mother tongue)
English (fluent, level C1)
Spanish (basic, level A2)

Education

Sep. 2023 – 2026 **PhD in Astronomy and Cosmic Physics**
At Institute for Theoretical Astrophysics (ITA), ZAH, Heidelberg University in Heidelberg, Germany
Supervisor: Prof. Michela Mapelli - Thesis title: "Binary Black Hole population in the Local Universe"

Oct. 2020 – Sep. 2022 **Master degree in Astrophysics**
At Università degli Studi di Padova in Padua (PD), Italy
Final grade: 110/110 *cum laude*, GPA: 29.4/30
Supervisor: Prof. Michela Mapelli - Thesis title: "Hierarchical Black Holes Mergers in Active Galactic Nuclei"

Sep. 2019 – Jul. 2020 **Master course in Matter Physics**
At Università di Pisa in Pisa (PI), Italy - GPA: 28.8/30

Sep. 2016 – Sep. 2019 **Bachelor degree in Physics**
At Università di Pisa in Pisa (PI), Italy
Final grade: 110/110 *cum laude*, GPA: 28.6/30
Supervisor: Prof. Pier Giorgio Prada Moroni - Thesis title: "Cooling evolution of white dwarfs and cosmochronology"

Sep. 2011 – Jul. 2016 **High school diploma**
At Liceo Scientifico Alfano da Termoli in Termoli (CB), Italy
Final grade: 100/100 *cum laude*

Aug. 2014 – Jul. 2015 **High school exchange year**
At Mead High School in Mead, Colorado, USA
GPA: 4.0/4 - ACT score: 33/36

Funding

February 1st – August 31st, 2023 **Pre-doctoral fellowship**, DEMOBLACK ERC Team, Research project titled "Hierarchical evolution of stellar-sized binary black holes in active galactic nuclei" under the supervision of Prof. Michela Mapelli at Università degli Studi di Padova, Padua (PD), Italy.

October 1st – December 31st 2022 **Internship** in Gravitational Waves Physics at IFAE-UAB, Barcelona, Spain under the supervision of Dr. Lluïsa-Maria Mir i Martínez

Current activity

I am a computational astrophysicist, working in the field of gravitational-wave astronomy and black hole dynamics. I am the main developer of a semi-analytical code for the active galactic nucleus (AGN) disk scenario for hierarchical black hole mergers.

Talks

December 7th – 15th, 2023 Talk titled "The impact of Gas Hardening on Hierarchical Black Hole Mergers in Migration Traps of Active Galactic Nuclei Disks" and discussion panel "Many Body Systems" at **RESCEU-NBIA workshop on GW sources** hosted by University of Tokyo in Tokyo, Japan

July 10th – 14th, 2023 Flash talk titled "Semi-analytical Model for Hierarchical Binary Black Hole Mergers" at "**GwPopNext**" workshop hosted by University of Milano Bicocca in Milan, Italy

Invited talks

June 26th – 30th, 2023 Invited poster presentation, workshop "**The Renaissance of Stellar Black-Hole Detections in The Local Group**" hosted by Lorentz Center in Leiden, Netherlands

April 19th – 21st, 2023 Invited Speaker, "**Physics of Data Workshop**", invited talk about "How do black holes grow? Active galactic nuclei as a breeding ground for intermediate-mass black holes", Venezia, Italy

Publications submitted

- **Impact of gas hardening on the population properties of hierarchical black hole mergers in AGN disks**, Maria Paola Vaccaro, Michela Mapelli, Carole Périgois, Dario Barone, M. Celeste Artale, Marco Dall'Amico, Giuliano Iorio, Stefano Torniamenti, 2023, arXiv:2311.18548
- **GW190521: a binary black hole merger inside an active galactic nucleus?**, Sophia Morton, Stefano Rinaldi, Alejandro Torres-Orjuela, Andrea Derdzinski, Maria Paola Vaccaro, Walter Del Pozzo, 2023, *Physical Review D.*, arXiv:2310.16025
- **The Hitchhiker's guide to the galaxy catalog approach for gravitational wave cosmology**, Jonathan R. Gair, Archisman Ghosh, Rachel Gray, ..., Maria Paola Vaccaro et al., 2022, *ApJ*, arXiv:2212.08694

Courses and schools

- 4th G2Net Training School "A network for Gravitational Waves, Geophysics and Machine Learning" hosted by Aristotle University of Thessaloniki (remote attendance) - March 28th-30th, 2023
- Python Course (level advanced) hosted by ICFO in Barcelona, Spain - November 24th-30th, 2022
- GW Open Data Workshop hosted by GWOSC (online event) - May 23rd-25th, 2022
- IPAM Workshop III: Source inference and parameter estimation in Gravitational Wave Astronomy (remote attendance) - November 15th-19th, 2021

Outreach

- Member of the cultural association Scholé based in Roccella Jonica (RC), Italy since 2020
- Member of the organizing committee for the Scholé "School of physics" for highschoolers since 2023
- Speaker and staff member at the 2023 Scholé Summer School titled "PHYSIS" ("Nature"). Talks:
  - "Understanding Nature with Quantum Mechanics" with Prof. Giancarlo Cella (Pisa, Italy) – July 25th, 2023
  - "The concept of space from Newton to Einstein" with Prof. Paolo Bussotti (Udine, Italy) – July 29th, 2023